

Yaswanth Ala

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Dallas, TX (Open to Relocation)

SUMMARY

AI Engineer with 4 years across software, data, and ML engineering, including ~2 years building production LLM and agentic systems. Built and deployed a procurement intelligence platform ingesting from 55+ government and enterprise portals, and shipped a cross-platform AI health app to Google Play solo. Owns the full path from retrieval and evaluation design through FastAPI services and containerized deployment on Azure and AWS.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript/TypeScript, Bash

Agentic AI & LLM Apps: LangGraph, LangChain, MCP (Model Context Protocol), Multi-Agent Orchestration, Tool/Function Calling (I/O Schemas, Guardrails, Structured Outputs), RAG, Hybrid Search (BM25 + Dense), Vector DBs (pgvector, Pinecone)

LLMs & Fine-Tuning: PyTorch, HuggingFace Transformers, OpenAI, Anthropic, Google Gemini, Llama, Mistral, Prompt Engineering, LoRA/QLoRA, Eval Datasets

LLM Evaluation & Reliability: Golden Datasets, Regression Tests, LLM-as-Judge Scoring, Scoring Rubrics, Quality Thresholds, Non-Deterministic Output Eval, Rate Limiting, Retries/Backoff, Idempotency, Replay/Backfill

ML, CV & Data Science: scikit-learn, XGBoost, Pandas, NumPy, OpenCV, TensorFlow/TensorFlow Lite, CNNs, Facial-Landmark Detection, Edge Deployment (Raspberry Pi), Vision-Model OCR, Feature Engineering, Anomaly Detection, NLP, Text Embeddings

Data & Backend: PySpark/SparkSQL, ETL/ELT, Data Modeling, Star Schemas, PostgreSQL, MongoDB, Redis, FastAPI, Django, Flask, React, REST APIs, JWT/OAuth 2.0, RBAC

Cloud & Ops: Azure (Azure OpenAI, Azure ML, Blob Storage), AWS (Redshift, EMR, Glue, Lambda, S3, EC2), Docker, Docker Compose, Nginx, GitHub Actions, MLflow, Weights & Biases, Structured Logging, Token/Cost Tracking

PROFESSIONAL EXPERIENCE

AI/ML Engineer

Dec 2024 – Present

EITACIES Inc.

Remote

- Built and deployed the RFP GenAI platform end to end — a production Python system that scrapes, normalizes, scores, and tracks solicitations from **55+ US and Canadian government and enterprise procurement portals** (SAM.gov, Bonfire, PlanetBids, CanadaBuys, LA County, TIPS USA, Cal eProcure), serving two client relevance profiles and cutting manual bid review time by ~70%.
- Engineered a deterministic two-tier relevance scoring engine — a client-profile scorer across five weighted dimensions (AI/ML core 38%, cloud & data platform 22%, consulting fit 20%, contract vehicle 12%, agency/sector 8%) with 3× title-match weighting, plus category base scores across 17 RFP categories with keyword boosts and non-IT vetoes — chosen over model inference so every categorization is explainable and auditable after early runs mislabeled non-technical solicitations as AI/ML.
- Built the concurrent scraping engine: per-portal Python modules orchestrated with `asyncio` and launched as background subprocesses so the API never blocks, combining requests against internal portal JSON APIs with Playwright headless Chromium for JavaScript-heavy and ASP.NET WebForms portals — with fast (1–2 min) and full (10–20 min) pipeline modes.
- Implemented idempotent ingestion against a JSONB-backed PostgreSQL schema (`tenders / seen_ids / pipeline_runs`): upsert on `solicitation_id` with COALESCE field merging, a dedup table keyed `UNIQUE(portal, rfp_id)`, auto-close of expired records, and idempotent startup migrations that evolve the schema without a migration framework.
- Shipped the operator-facing dashboard (Alpine.js, vanilla CSS, no build step) with a searchable score-sorted tender grid, pipeline trigger with run history, and a Kanban lifecycle tracker (New → Identified → Active → In Progress) synced to Attio CRM; deployed via Docker Compose on DigitalOcean behind Nginx with HTTPS, and authored the operations runbook and security-hardening checklist for client handover.

Environment: Python, FastAPI, uvicorn, Alpine.js, PostgreSQL/Supabase, asyncio, Playwright, Docker Compose, DigitalOcean, Nginx, Attio CRM

AI Engineer Intern

May 2024 – Aug 2024

Thomson Reuters

Remote

- Built the ONESOURCE Intelligent Assistant, a RAG chatbot answering customer queries about Thomson Reuters tax products, grounding every response in product documentation and tax knowledge bases with inline source citations to reduce hallucination on compliance-sensitive answers.
- Developed the document chunking and ingestion pipeline — document-type-aware splitting, metadata enrichment, and filtering — across a heterogeneous corpus of manuals, release notes, and knowledge-base articles to improve answer grounding on edge-case tax queries.
- Served the assistant through a FastAPI service layer over Azure OpenAI, with guardrails for out-of-scope and refusal handling, prompt versioning, and structured request/response logging; containerized with Docker.
- Established a golden Q&A evaluation set with product and support stakeholders to score answer grounding and citation correctness, catching retrieval regressions before release rather than in customer-facing testing.

Environment: Python, FastAPI, Azure OpenAI, RAG/Vector Retrieval, Document Chunking, Prompt Versioning, Docker

Systems Engineer

Infosys

Jun 2021 – Jan 2022

India

- Led an end-to-end legacy database migration to AWS Redshift and EMR, using S3 and Lambda to build scalable data warehouse solutions and data models supporting analytics and downstream ML workloads.
- Designed PySpark ETL pipelines on EMR processing **TB-scale datasets**, applying partitioning, predicate pushdown, and broadcast-join optimizations to cut job runtimes and cluster costs.
- Built automated data validation and monitoring frameworks in AWS Glue and Python with statistical anomaly detection and isolation-forest outlier flagging, reducing data inconsistencies and validation runtime.
- Engineered dimensional and star-schema data models in Redshift to enable analyst self-service, defining grain, slowly-changing dimensions, and aggregation strategy with business stakeholders.

Environment: Python, PySpark, SQL, AWS Redshift, EMR, S3, Glue, Lambda, scikit-learn, Star-Schema Modeling

Full Stack Developer

Nspira Management Services

Jul 2019 – Jun 2021

India

- Developed full-stack online course platforms in Django, React, and MongoDB serving **50,000+ active learners**, with personalized course recommendations from collaborative filtering and content-based ranking models built in scikit-learn and PySpark.
- Built scalable RESTful APIs on AWS EC2 with Nginx and Gunicorn handling high concurrent load across enrollment, video streaming, assessment, and progress-tracking services.
- Implemented multi-tier caching with Redis for hot data and CloudFront/S3 for static and video assets, improving page load times for end users.
- Secured the platform with JWT, OAuth 2.0, and RBAC across instructor, student, and administrator roles; established Jenkins/Docker CI/CD reducing release cycles from weeks to days.

Environment: Python, Django, React, MongoDB, Redis, AWS EC2/S3/CloudFront, Nginx, Gunicorn, Jenkins, Docker

EDUCATION

Wichita State University

Master of Science in Computer Science

Jan 2022 – Dec 2023

Wichita, KS

Amrita School of Engineering

Bachelor of Technology in Computer Science

[Start] – [End]

India

CERTIFICATIONS

- AWS Certified Data Engineer – Associate

Nov 2024
- Microsoft Certified: Azure AI Engineer Associate (AI-102)

Apr 2024
- Microsoft Certified: Azure AI Fundamentals (AI-900)

Feb 2024
- Palantir Foundry & AIP Builder Foundations

[Date]

FEATURED PROJECTS

FitNova — AI Health-Tracking App

2026

Environment: React 19, TypeScript, Capacitor, Firebase, Gemini Vision, Cloudflare Workers | [GitHub](#)

- Solo-designed and shipped a production cross-platform (Android / iOS / web) health app to Google Play from a single React 19 / TypeScript / Vite codebase, using Capacitor to build native Kotlin (Android) and Swift (iOS) shells — including a custom Kotlin Health Connect plugin for sleep-stage data — in ~9 days.
- Built an AI ingredient-label scanner that captures packaging images and transcribes ingredient lists via Google Gemini vision (OCR), with a deliberate human-in-the-loop review/correct step before rule-based health scoring — prioritizing accuracy over blind photo-to-verdict inference.
- Hardened the architecture in a self-driven security review: moved the Gemini API key out of the client bundle into a Cloudflare Worker proxy that verifies Firebase ID tokens, and enforced owner-only Firestore rules for zero cross-user data exposure by design.
- Integrated Health Connect (steps, heart rate, workouts, sleep across Samsung Health / Google Fit) and Open Food Facts; implemented Mifflin–St Jeor BMR/TDEE targets and adaptive meal reminders learned from logged eating times.

Intelligent Document Assistant (intelli-doc)

2025 – 2026

Environment: React 19, TypeScript, Vite, Google Gemini, pdfjs-dist | [GitHub](#)

- Built a browser-native document question-answering assistant — client-side PDF parsing with pdfjs-dist, in-memory embedding and retrieval, and grounded answering via the Google Gemini API — with **zero server-side document storage**, so sensitive documents never leave the user’s machine.
- Designed chunking and retrieval to run entirely in-browser under memory constraints, trading infrastructure for a privacy-preserving architecture with no backend to operate.
- Currently extending the system with a FastAPI + LangGraph backend over pgvector to support larger corpora, cross-encoder re-ranking, and an LLM-as-judge verification pass that re-runs retrieval when a response falls below a groundedness threshold.

Real-Time Drowsiness Detection System

2020

Environment: Python, OpenCV, TensorFlow Lite, CNNs, Raspberry Pi, IoT | [GitHub](#)

- Developed a real-time computer vision system detecting driver drowsiness from live video, combining facial-landmark detection with a CNN classifier over extracted eye and face regions to reach **92% detection accuracy**, optimized with TensorFlow Lite and model compression to hold real-time inference on Raspberry Pi.
- Cut false alarms by **35%** through temporal feature extraction over landmark sequences, scoring sustained closure patterns across frames instead of firing on single-frame predictions.